

Xiaochuan Li

M.Sc. Human and AI · University of Technology Nuremberg

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RESEARCH INTERESTS

Experimental and computational semantics; epistemic modality; discourse update; language and reasoning; linguistically informed evaluation of language models.

EDUCATION

M.Sc., Human and Artificial Intelligence 2025 – 2027

University of Technology Nuremberg (UTN)

Coursework: cognitive psychology, experimental design and statistics, interpretability, deep learning, epistemology.

German language program 2024 – 2025

University of Marburg · intensive German for graduate study

B.A., Philosophy 2020 – 2024

Shenzhen University

Senior thesis on Leibniz's *characteristica universalis* (advisor: Zang Yong). Coursework in philosophy of language, analytic philosophy, logic, and mathematical logic.

MANUSCRIPTS AND RESEARCH PROJECTS

Epistemic modality and the maintenance of alternatives — manuscript in preparation

Develops diagnostics for distinguishing content-specific alternative maintenance from mention, generic uncertainty, and perseveration. Construct brief under review by a formal semanticist.

M.Sc. thesis: learnability of modal discourse effects — in design

Tests whether a small language model can acquire the target distinctions from developmentally realistic linguistic input and which input patterns are necessary.

Cross-Lingual Prototypicality Bias in Multimodal Evaluation Metrics and VLM Judges — first-author; in revision for arXiv

Controlled cross-lingual evaluation across seven languages and two VLM families; about 12,600 judgments, item-clustered analysis, a human control, and public code. [GitHub](#) ↗

MODUS — exploratory methods pilot

Developed controlled contrasts and interventions for separating modal vocabulary from certainty-related behaviour; identified lexical, base-rate, and generic-disruption confounds.

RESEARCH EXPERIENCE

Latent Control States — ongoing collaboration with DFKI

Designed a utilitarian minimal-pair experiment varying dilemma, benefit size, and framing to test whether prompt framing changes a causally relevant internal state.

METHODS AND TECHNICAL SKILLS

Programming and data. Python, NumPy, pandas, scikit-learn, Git, LaTeX.

ML / NLP. PyTorch, Hugging Face Transformers, model inference, dataset pipelines, LoRA fine-tuning.

Interpretability. Linear and structural probing, activation patching, steering, ablation (working proficiency).

Experimental design and statistics. Controlled stimuli, minimal pairs, preregistration, mixed-effects analysis (working proficiency).

LANGUAGES

Mandarin Chinese (native), German (C1), English (C1), Japanese (learning).